BMAT201L- Complex Variables and Linear Algebra-Practice Questions 1

1. Discuss the singularities of (i) $f(z)=\dfrac{e^z}{z^2+1}$ (ii) $f(z)=\dfrac{1-\cos z}{z}$ (iii) $f(z)=e^{1/z}$

   (iv) $f(z)=\dfrac{z^2}{(z-1)^2(z+2)}$.

   Answer: (i) $z=-i,\ i$ are isolated singular points of $f(z)=\dfrac{e^z}{z^2+1}$

   (ii) $z=0$ is a removable singularity of $f(z)=\dfrac{1-\cos z}{z}$

   (iii) $z=0$ is an essential singularity of $f(z)=e^{1/z}$

   (iv) $z=1,\ -2$ are the singular points of $f(z)$. Here $z=1$ is a pole of order 2 and $z=-2$ is a simple pole..

2. Find the poles and the corresponding residues of the function $f(z)=\dfrac{z^2}{(z-1)(z-2)^2}$

   Answer: Here $z=1$ is a pole and $z=2$ is a pole of order 2. And the corresponding Residues are $1,\ 0$ respectively.

3. Evaluate $\int_C \dfrac{z+4}{z^2+2z+5}\,dz$ where C is the circle $|z|=1$.

   Answer: Clearly $f(z)=\dfrac{z+4}{z^2+2z+5}$ is analytic at all the points within and on the circle $|z|=1$.

   Therefore, by Cauchy's theorem, we have $\int_C \dfrac{z+4}{z^2+2z+5}\,dz=0$.

4. Evaluate $\int_C \dfrac{z^3e^{-z}}{(z-1)^2}\,dz$ where C is $|z-1|=\dfrac{1}{2}$.

   Answer: By Cauchy's integral formula, $\int_C \dfrac{z^3e^{-z}}{(z-1)^2}\,dz=\dfrac{\pi i}{e}$

5. Evaluate $\int_c \dfrac{z^2+2z}{(z+1)^2(z^2+4)}\,dz$ where C is $|z|=\dfrac{3}{2}$ using Residue theorem.

   Answer: Here $z=-1$ is a pole of order two which is inside of C and $z=-2i,\ z=2i$ are simple poles which are outside of C. Therefore $\int_c \dfrac{z^2+2z}{(z+1)^2(z^2+4)}\,dz=2\pi i(\operatorname{Res}.f(z))_{z=-1}=2\pi i\left(\dfrac{2}{25}\right)$

6. Evaluate $\int_C \frac{4-3z}{z(z-1)(z-2)}\,dz$ where C is $|z| = \frac{3}{2}$ using Residue theorem.

Answer: $\int_C \frac{4-3z}{z(z-1)(z-2)}\,dz = 2\pi i$.

7. Evaluate $\int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos\theta}\,d\theta$ using Residue theorem. **Answer:** $\frac{\pi}{6}$

8. Evaluate $\int_0^{\infty} \frac{2x^2-1}{x^4+5x^2+4}\,dx$ using Residue theorem. **Answer:** $\frac{\pi}{4}$

9. Evaluate $\int_0^{\infty} \frac{x\sin x}{1+x^2}\,dx$ using Residue theorem. **Answer:** $\frac{\pi i}{2e}$

10. Evaluate $\int_{-\infty}^{\infty} \frac{\cos 2x}{x}\,dx$. **Answer:** 0